

JEE MAIN REVISION MASTER: QUADRATIC EQUATION AND INEQUALITIES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 196 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Roots of Quadratic & Vieta's Relations	132 PYQs	Quadratic Roots
Very High	Location of Roots & Quadratic Inequalities	67 PYQs	Location of Roots

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Roots of Quadratic & Vieta's Relations	$\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$, $ \alpha - \beta = \frac{\sqrt{D}}{ a }$, $aS_n + bS_{n-1} + cS_{n-2} = 0$ (Newton)
Location of Roots & Quadratic Inequalities	$\text{Both roots } > k \iff D \geq 0, -\frac{b}{2a} > k, a \cdot f(k) > 0$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Roots of Quadratic & Vieta's Relations (Very High Yield — 132 PYQs)

Core Principle: Discriminant analysis for real/complex roots, symmetric root functions, and Newton's power sum theorem.

■ **Caution / JEE Pitfall:** If coefficients a, b, c are rational and D is not a perfect square, roots occur in irrational conjugate pairs $(p \pm \sqrt{q})$.

■ **Speed Trick:** Newton's theorem: If α, β are roots of $ax^2 + bx + c = 0$ and $S_n = \alpha^n + \beta^n$, then $aS_n + bS_{n-1} + cS_{n-2} = 0$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\lim_{x \rightarrow 2} \frac{\tan(x-2)}{(x^2+(p-2)x-2p)(x-2)^2} = 5$ for some $r, p \in \mathbb{R}$. If the set of all possible values of p is $\{r\}$, then $r =$ _____" [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "Let $\alpha, \alpha+2, \alpha \in \mathbb{Z}$, be the roots of the quadratic equation $x(x+2)+(x+1)(x+3)+(x+2)(x+4)+\dots+(x+n-1)(x+n+1)=4n$ for some $n \in \mathbb{N}$." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "Let α, β be the roots of the quadratic equation $12x^2-20x+3\lambda=0, \lambda \in \mathbb{Z}$. If $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{3}{2}$, then $|\alpha - \beta| =$ _____" [Ans: N/A]

3.2 Location of Roots & Quadratic Inequalities (Very High Yield — 67 PYQs)

Core Principle: Conditions for roots relative to real numbers k , k_1 , k_2 , and signs of quadratic expression $ax^2 + bx + c$.

■ **Caution / JEE Pitfall:** For both roots to be greater than k , checking $a \cdot f(k) > 0$ alone is NOT sufficient; you must ALSO check $D \geq 0$ and $-(b/2a) > k$.

■ **Speed Trick:** If $a > 0$, the quadratic $f(x) = ax^2 + bx + c > 0$ for all real x if and only if $D < 0$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\lim_{x \rightarrow 2} \frac{\tan(x-2)(rx^2+(p-2)x-2p)}{(x-2)^2} = 5$ for some $r, p \in \mathbb{R}$. If the set of all possible values of q , such that the roots of the equation $rx^2 - px + q = 0$ lie in $(0, 2)$, be the interval $(\alpha, \beta]$, then $4(\alpha + \beta)$ equals : [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "The number of the real solutions of the equation: $x|x+3|+|x-1|-2=0$ is..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "If the quadratic equation $(\lambda+2)x^2-3\lambda x+4\lambda=0, \lambda \neq -2$, has two positive roots, then the number of possible integer values of λ is..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

Let $\lim_{x \rightarrow 2} \frac{\tan(x-2)(rx^2+(p-2)x-2p)}{(x-2)^2} = 5$ for some $r, p \in \mathbb{R}$. If the set of all possible values of q , such that the roots of the equation $rx^2 - px + q = 0$ lie in $(0, 2)$, be the interval $(\alpha, \beta]$, then $4(\alpha + \beta)$ equals :

- (A) 11
- (B) 13
- (C) 17 [Correct]
- (D) 21

Correct Answer: C

Solution Analysis: We are given $\lim_{x \rightarrow 2} \frac{\tan(x-2)(rx^2+(p-2)x-2p)}{(x-2)^2} = 5$. We have to first find the relation between r and p , and then determine all values of q for which the roots of $rx^2 - px + q = 0$ lie in $(0, 2)$. Step 1: Simplify the limit. Let $f(x) = rx^2 + (p-2)x - 2p$. Then the given limit becomes $\lim_{x \rightarrow 2} \frac{\tan(x-2)f(x)}{(x-2)^2} = 5$. Now, as $x \rightarrow 2$, $\tan(x-2) \sim x-2$. So the limit becomes $\lim_{x \rightarrow 2} \frac{(x-2)f(x)}{(x-2)^2} = \lim_{x \rightarrow 2} \frac{f(x)}{x-2} = 5$. For this limit to be finite,

Benchmark Problem 2: JEE Main 2026 (Online) 4th April Morning Shift (Q#3)

If the set of all solutions of $|x^2+x-9|=|x|+|x^2-9|$ is $[\alpha, \beta] \cup [\gamma, \infty)$, then $(\alpha^2 + \beta^2 + \gamma^2)$ is equal to:

- (A) 9
- (B) 18 [Correct]
- (C) 36
- (D) 72

Correct Answer: B

Solution Analysis: We need to solve $|x^2+x-9|=|x|+|x^2-9|$. We will do it by considering intervals where the expressions inside modulus change sign. The critical points are obtained from: $x=0$ from $|x|$, $x=\pm 3$ from $|x^2-9|$, roots of $x^2+x-9=0$, but for NCERT-style interval splitting here, it is enough to use the main sign-change points $-3, 0, 3$ and then check sign of x^2+x-9 in each interval. Now solve interval-wise. 1. For $x \leq -3$ Here, $|x|=-x$, $|x^2-9|=x^2-9$. Also $x^2+x-9 > 0$, so $|x^2+x-9|=x^2+x-9$. Thus, $x^2+x-9=-x+(x^2-9)$ $x^2+x-9=x^2-x-9$ $x=-x$ $2x=0$ $x=0$ (Not in interval).